



**REPUBLIC OF TURKEY
YILDIZ TECHNICAL UNIVERSITY
FACULTY OF ELECTRICAL-ELECTRONICS ENGINEERING
DEPARTMENT OF CONTROL AND AUTOMATION ENGINEERING**

GRADUATION PROJECT

**Design and Implementation of a 2-DOF Shaking Table for Simulating Seismic
Earthquake Motions with MRAC Controller and Classical Controller Models**

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Istanbul, 2025

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YILDIZ TECHNICAL UNIVERSITY
DEPARTMENT OF CONTROL AND AUTOMATION ENGINEERING

Design and Implementation of a 2-DOF Shaking Table for Simulating Seismic Earthquake Motions with MRAC Controller and Classical Controller Models

A graduation project submitted by Aylin ÖZTÜRK, Çağla ÜZÜMCÜ, Eren Buğra GÜNÜLTAŞ in partial fulfillment of the requirements for the degree of bachelor is approved by the committee on 13.06.2025 in Department of Control and Automation Engineering, Faculty of Electrical and Electronics Engineering.

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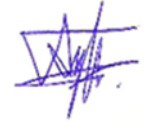
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ÖNSÖZ

Bu bitirme çalışması, mühendislik eğitimimiz süresince edindiğimiz teorik ve pratik bilgileri birleştirerek deprem simülasyonu ve kontrol sistemleri alanında katkı sağlamak amacıyla gerçekleştirilmiştir.

" MRAC Kontrol Metodu ve Klasik Kontrol Metodları ile Sismik Deprem Hareketlerini Simüle Etmek için 2-DOF Sarsma Masasının Tasarımı ve Uygulanması " başlıklı projemizde sağladığı destekten dolayı TÜBİTAK'a teşekkür ederiz. Ayrıca, bilgi ve tecrübeleriyle bize yol gösteren değerli danışmanımız Dr. Öğr. Üyesi Claudia Fernanda YAŞAR'a ve süreç boyunca bize destek olan ailelerimize ve arkadaşlarımıza minnettarız.

Bu çalışmanın, ülkemizde yapı güvenliği ve deprem farkındalığına katkı sağlamasını temenni ederiz.

Haziran, 2025

Aylin ÖZTÜRK, Çağla ÜZÜMCÜ, Eren Buğra GÜNÜLTAŞ

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This graduation project was carried out to combine the theoretical and practical knowledge acquired during our engineering education and to contribute to the fields of earthquake simulation and control systems.

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We hope that this study will contribute to structural safety and raise awareness of earthquake preparedness in our country.

June, 2025

Aylin ÖZTÜRK, Çağla ÜZÜMCÜ, Eren Buğra GÜNÜLTAŞ

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LIST OF SYMBOLS

Γ	Motor torque
K	Motor constant
v	Viscous friction
J	Motor inertia
V	Voltage
F	Force
η	Ball Screw efficiency
T	Torque
p	Ball Screw pitch

LIST OF ABBREVIATIONS

PID	Proportional Integrator Derivative
PIV	Proportional Integrator Velocity
MRAC	Model Reference Adaptive Controller
DOF	Degree of Freedom
DC	Direct Current

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Design and Implementation of a 2-DOF Shaking Table for Simulating Seismic Earthquake Motions with MRAC Controller and Classical Controller Models

Aylin ÖZTÜRK, Çağla ÜZÜMCÜ and Eren Buğra GÜNÜLTAŞ

Department of Control and Automation Engineering

Graduation Project

Advisor: Asst. Prof. Dr. Claudia Fernanda YAŞAR

This project aims to enhance structural safety in earthquake-prone regions such as Turkey by developing a two-degree-of-freedom (2-DOF) shaking table system. The system simulates realistic seismic motions, allowing for the dynamic testing and evaluation of structures and construction materials. This contributes significantly to the advancement of earthquake-resistant building technologies.

The shaking table is mechanically designed to move along two horizontal axes and is powered by DC motors, which are precisely controlled using two types of control algorithms: the classical Proportional-Integral-Derivative (PID) controller and the Model Reference Adaptive Control (MRAC) algorithm. These control methods are modeled and tested in MATLAB Simulink to analyze their effectiveness under variable seismic conditions. The MRAC approach demonstrates superior adaptability and precision in handling non-linear and time-varying system dynamics compared to the PID controller.

The project involves several technical phases including the mechanical design of the shaking platform, the development of the DC motor driver circuit, and the implementation of both control strategies. These phases are structured into work packages with clear timelines and deliverables. Simulation-based testing will be conducted to assess the system's dynamic performance and control accuracy under seismic input signals.

Beyond its technical scope, the project contributes to scientific understanding in control engineering by offering a comparative analysis of adaptive and classical control

systems in earthquake simulation. From an engineering perspective, it provides valuable insights into the development of multi-axis shaking tables. On a societal level, it helps raise awareness about earthquake preparedness and building safety, providing a foundation for future research and educational outreach in disaster-resilient design.

This work demonstrates how modern control techniques, such as MRAC, can be effectively applied to critical engineering problems like seismic simulation, ultimately supporting the creation of safer, smarter infrastructure in vulnerable regions.

Key words: Adaptive Control (MRAC), PID Control, Control Systems, Dynamic Systems, Earthquake Engineering

YILDIZ TECHNICAL UNIVERSITY
DEPARTMENT OF CONTROL AND AUTOMATION ENGINEERING

**MRAC Kontrol Methodu ve Klasik Kontrol Methodları ile Sismik Deprem Hareketlerini
Simüle Etmek için 2-DOF Sarsma Masasının Tasarımı ve Uygulanması**

Aylin ÖZTÜRK, Çağla ÜZÜMCÜ, Eren Buğra GÜNÜLTAŞ

Kontrol ve Otomasyon Mühendisliği Bölümü

Bitirme Çalışması

Danışman: Asst. Prof. Dr. Claudia Fernanda YAŞAR

Bu proje, Türkiye gibi deprem riski yüksek ülkelerde yapıların güvenliğini artırmaya yönelik olarak, iki serbestlik dereceli (2-DOF) bir sarsma masası sisteminin geliştirilmesini amaçlamaktadır. Deprem testleri için oldukça önemli olan bu sistem, yapıların gerçekçi sismik yükler altında nasıl davrandığını simüle edebilmekte ve böylece yapı tasarımı ve test süreçlerinin etkinliğini artırmaktadır.

Sistem, yatay eksenlerde hareket edebilen mekanik bir sarsma masası yapısına ve bu yapıyı kontrol eden DC motorlara sahiptir. Kontrol sistemi, hem klasik PID (Oransal-Türevsel-İntegral) algoritması hem de Model Referanslı Uyarlamalı Kontrol (MRAC) yöntemi ile gerçekleştirilmiştir. Bu iki kontrol yaklaşımı MATLAB Simulink ortamında modellenip benzetilerek karşılaştırılmış, MRAC'nin değişen sismik girdilere daha hızlı ve doğru yanıt verdiği gözlemlenmiştir. MRAC, sistem parametreleri değişse bile performansını koruyabilen bir yapıya sahiptir.

Proje süreci, mekanik sistemin tasarımı, DC motor sürücüsünün geliştirilmesi, kontrol algoritmalarının yazılım ortamında modellenmesi ve uygulamaya alınması gibi aşamaları içermektedir. Her bir aşama belirli zaman çizelgeleri ve hedeflerle tanımlanmış iş paketlerine ayrılmıştır. Gerçekleştirilecek testlerle sistemin hem dinamik davranışı hem de kontrol kararlılığı değerlendirilecektir.

Bu çalışma, sadece mühendislik açısından değil, aynı zamanda bilimsel ve toplumsal açıdan da önemli katkılar sağlamaktadır. Bilimsel olarak kontrol mühendisliği alanına MRAC ve PID sistemlerinin deprem simülasyonundaki etkinliğini kıyaslayan bir yaklaşım sunarken; mühendislik açısından sarsma masası gibi test sistemlerinin tasarımı konusunda örnek bir proje oluşturmaktadır. Toplumsal düzeyde ise yapı güvenliğine ve deprem farkındalığına katkı sağlayacak bir altyapı sunmaktadır. Bu proje, ileri kontrol tekniklerinin doğal afetlere karşı dirençli yapıların geliştirilmesinde nasıl etkili kullanılabileceğini göstermektedir.

Anahtar Kelimeler: Uyarlamalı Kontrol (MRAC), PID Kontrol, Kontrol Sistemleri, Dinamik Sistemler, Deprem Mühendisliği

YILDIZ TEKNİK ÜNİVERSİTESİ
KONTROL VE OTOMASYON MÜHENDİSLİĞİ BÖLÜMÜ

INTRODUCTION

1.1 Literature Survey

The design and control of shaking tables have been a subject of extensive research over the past decades due to their critical role in simulating seismic activities and evaluating structural performance of buildings under earthquake loads. Early studies primarily focused on the mechanical design and 1-DOF systems [1]. There are studies explored the dynamic behavior of mechanical shaking tables, emphasizing their potential in replicating low-frequency earthquake signals [2]. However, these early systems often lacked the control precision and adaptability required for more complex motion patterns.

In the early stages of shaking table control, traditional linear control strategies, primarily consisting of PID (Proportional Integral Derivative) and three-parameter controllers, dominated the field. It is demonstrated that the efficacy of three-variable controllers, integrating displacement, velocity, and acceleration feedback to effectively broaden the bandwidth and reduce distortion in waveform tracking [3]. Complementary studies employed PID tuning—including proportional, integral, derivative, and delta-pressure gains—to characterize uniaxial hydraulic systems and optimize their frequency-domain responses [4]. However, despite improved performance under nominal conditions, these classical controllers struggled significantly when subjected to dynamic changes caused by placing test specimens on the table [5].

With advancements in control theory, the focus shifted towards the integration of classical control methods such as PID controllers to improve the accuracy of table motion tracking. Researchers demonstrated that PID-controlled systems could provide acceptable performance in ideal conditions, but their effectiveness decreased significantly when faced with system uncertainties and nonlinear behaviors [6].

To address these limitations, MRAC schemes have gained popularity in recent years. MRAC offers the advantage of adjusting controller parameters in real time based on the discrepancies between the system output and a desired reference model [7]. Applying MRAC frameworks to vibration and seismic simulations, studies show significant improvements in robustness and adaptability compared to fixed-gain controllers [8].

In terms of mechanical design, the development of 2-DOF shaking tables has provided a more realistic simulation environment by allowing both longitudinal and lateral motion. Studies implemented multi-degree-of-freedom tables with various control strategies, highlighting the need for advanced control techniques that can handle the increased complexity and dynamic coupling between axes [9].

Despite these advancements, a gap remains in the comparative evaluation of MRAC controllers and classical controllers on a 2-DOF shaking platform. Most existing literature either focuses on single-degree-of-freedom systems or evaluates one control strategy in isolation. Furthermore, physical implementation aspects—such as actuator response time, table resonance frequencies, and feedback system accuracy—are often underrepresented in simulation-heavy studies.

This study aims to bridge this gap by presenting the complete design and implementation of a 2-DOF shaking table system, equipped with both MRAC and classical control schemes. The system will be tested using recorded seismic inputs, and the control performance will be evaluated in terms of tracking accuracy, adaptability, and robustness under varying conditions.

1.2 Objective of Study

The primary objective of this study is to design, develop, and implement a 2-DOF shaking table capable of replicating realistic seismic ground motions. This platform aims to provide an effective means of testing structural systems and control strategies, particularly a Model Reference Adaptive Control (MRAC) scheme and classical controllers (e.g., PID), under earthquake-like excitations.

The motivation for this research stems from the significant damage caused by recent earthquakes in Türkiye. These seismic events have led to widespread structural failures, building collapses, and substantial loss of life and property, underscoring the urgent need for more robust earthquake-resistant designs and effective control strategies for structural systems [10]. Many structures, especially reinforced concrete buildings, have suffered severe damage due to inadequate seismic detailing, highlighting vulnerabilities in existing construction practices [11]. These challenges necessitate the development of experimental platforms that can accurately simulate seismic ground motions for in-depth analysis and controller validation.

Therefore, this study aims to design, construct, and experimentally validate a 2-DOF shaking table equipped with both classical (PID-based) and MRAC schemes. The main objectives are:

- Developing a 2-DOF shaking table that can reproduce the complex horizontal and vertical components of seismic ground motions.
- Implementing PID and MRAC controllers and comparing their performance metrics.
- Conducting seismic simulation tests using scaled earthquake records to quantify improvements in waveform fidelity under MRAC compared to classical control.
- Providing empirical evidence supporting MRAC's ability to enhance structural testing frameworks and contribute to the development of earthquake-resilient construction practices.

DESIGN AND IMPLEMENTATION OF THE SHAKING TABLE

2.1 Reduced DC Motor Modeling and Transfer Function Representation

A DC motor is a widely used actuator in control systems, known for its simplicity and ability to provide a linear relationship between input voltage and output speed under certain assumptions. To model the behavior of a DC motor, both its electrical and mechanical components are considered.

The DC motor reduced dynamics [12]:

$$\Gamma = KV = J\dot{\theta} + v\theta + \tau^{nlf} \quad (2.1)$$

Where:

Γ : Motor torque.

K : Motor constant.

v : Viscous friction.

J : Motor inertia.

τ^{nl} : Static Coulomb friction.

V : Input (voltage).

By applying the Laplace Transform (assuming zero initial conditions), the transfer function between the input voltage $V(s)$ and the output angular velocity $\omega(s)$ can be derived [13]:

$$KV(s) = Js^2\theta(s) + v\theta(s) \quad (2.2)$$

$$\frac{\theta(s)}{V(s)} = \frac{\frac{k}{J}}{s(s+\frac{v}{J})} = \frac{A}{s(s+B)} \quad (2.3)$$

$$A = \frac{k}{J} \quad B = \frac{v}{J} \quad (2.4)$$

2.2 State-space Representation of a DC Motor

The state-space representation of the DC motor is [13]:

$$KV = J\ddot{\theta} + v\dot{\theta} \quad (2.5)$$

The general form of a state-space model is:

$$\dot{x} = Ax + Bu \quad (2.6)$$

$$y = Cx + Du \quad (2.7)$$

where

A: State Matrix

B: Input Matrix

C: Output Matrix

D: Direct Feedthrough Matrix

By considering the state vector $x = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$ and $\dot{x} = \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \end{bmatrix}$ with the input being the voltage V and the output being the angular speed $\dot{\theta}$, the following state and output equations are obtained:

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{v}{j} \end{bmatrix} x + \begin{bmatrix} 0 \\ \frac{k}{j} \end{bmatrix} V(t) \quad (2.8)$$

$$y = [0 \quad 1]x + [0]V(t) \quad (2.9)$$

This state-space representation can be simulated in Simulink using the following diagram. But in this study, there is not any relationship between the input and the output, and therefore, $D=0$:

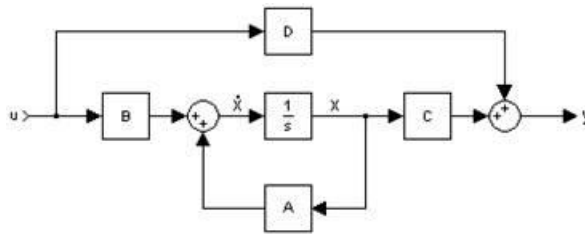


Figure 2.1: State-space model block diagram

2.3 Real-time Implementation

In the real-time implementation, the process starts with system identification based on experimental input-output data obtained from the physical system. Initially, rough estimates for the parameters A and B are assigned. Using the Parameter Estimation tool in Simulink, these parameters are iteratively tuned to minimize the error between

the simulated and measured outputs. Once accurate values of A and B are obtained, the resulting dynamic model becomes suitable for controller design. In this study, a Proportional-Integral-Velocity (PIV) control strategy is employed to regulate the system's behavior and ensure a fast and stable response under varying operating conditions.

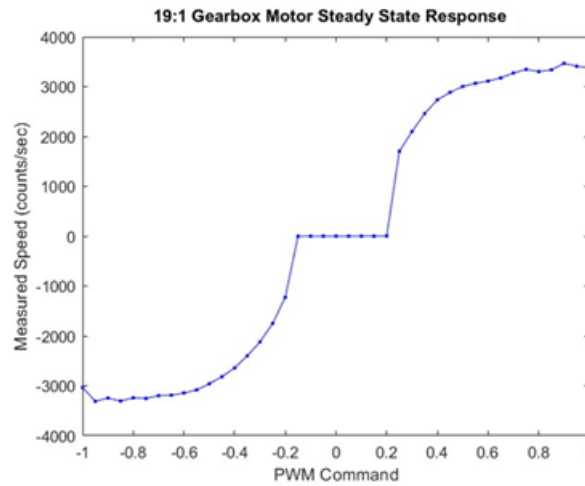


Figure 2.2: Motor Steady State Response

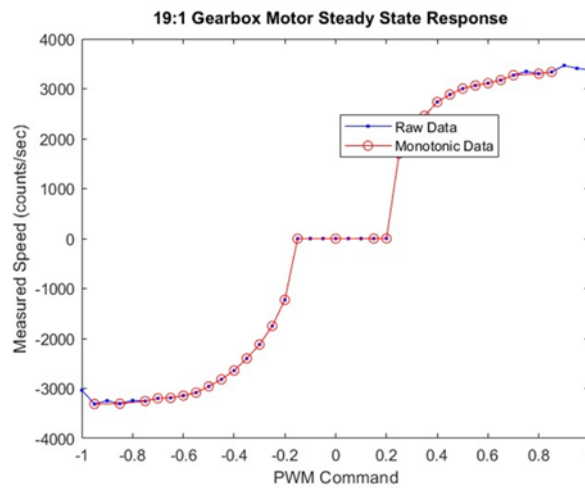


Figure 2.3: Motor Steady State Response-2

In the real-time Simulink model, a pulse generator is used to provide periodic reference speed commands. These commands are fed into a 1D Lookup Table block, which maps the desired speed values to appropriate PWM commands based on previously identified experimental data (PWMcmdMono and speedMono). The mapped PWM signal is then sent to the motor block, which interfaces with the physical hardware. The encoder measures the motor's position in real time, from which the actual speed is derived using a discrete differentiation block.

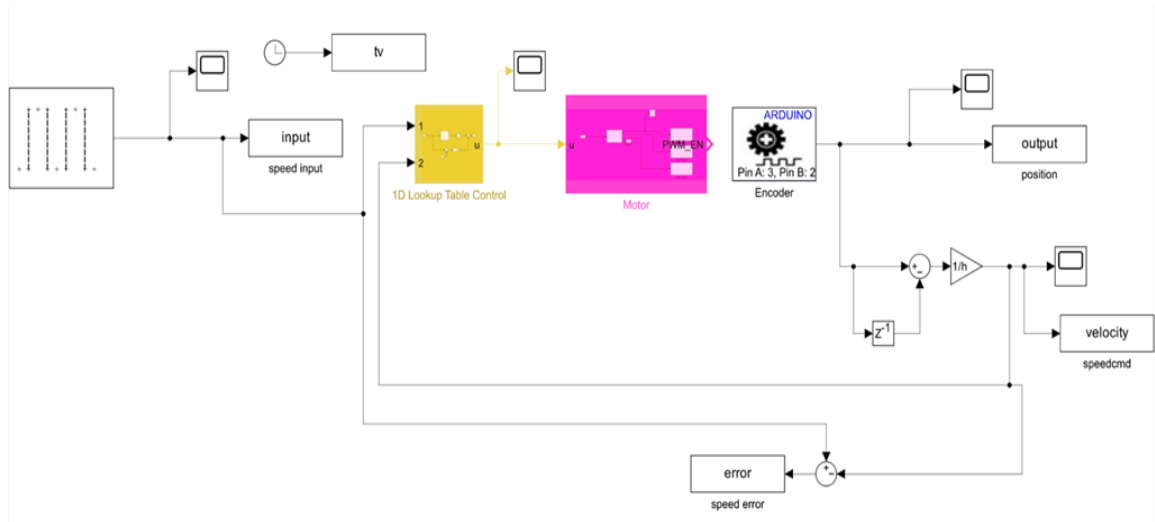


Figure 2.4: Real-Time Motor Speed Control Using 1D Lookup Table in Simulink

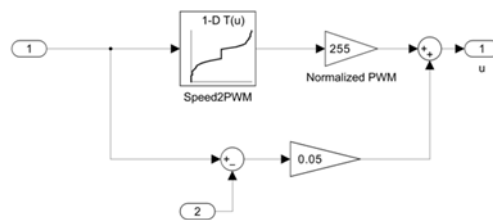


Figure 2.5: 1D Lookup Table Control

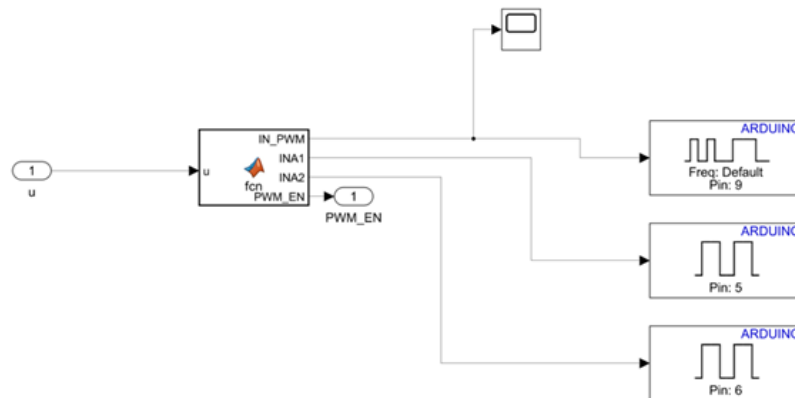


Figure 2.6: Motor

2.4 DC Motor Parameter Identification

The motor's parameters A and B were identified using Simulink's Parameter Estimator tool. The measured velocity and the input duration (tv) were used for comparison. A dead zone block (± 0.2) was added to account for motor nonlinearity. The estimation was done using the Nonlinear Least Squares method with:

- Max iterations: 100
- Parameter tolerance: 0.001

Accurate values of A and B were obtained for use in controller design.

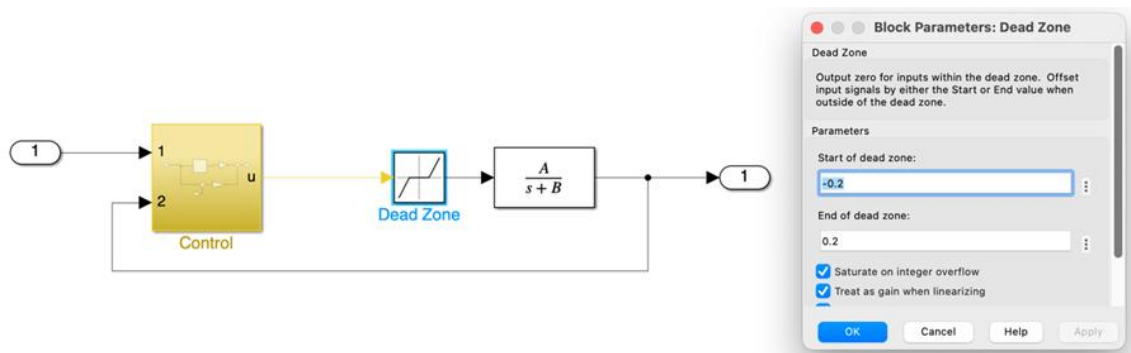


Figure 2.7: Parameter Estimation Model for A and B Using Simulink

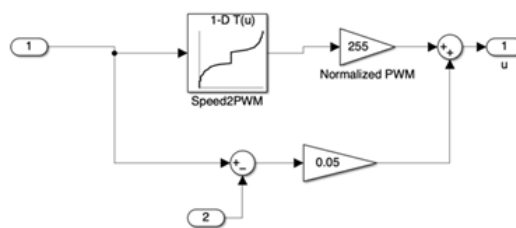


Figure 2.8: Control

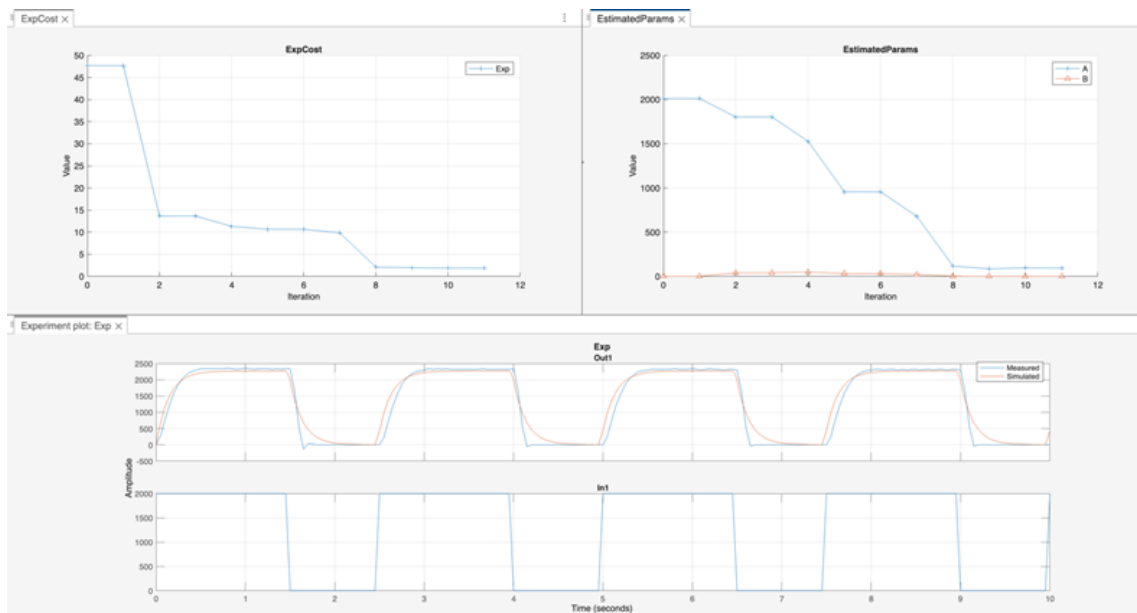


Figure 2.9: System Identification/ Parameter Estimation and Simulation Results

Parameter	Initial Value	Estimate
A	95.3222414923339	Yes
B	2.49510644698831	Yes

Figure 2.10: Estimated Parameter Values

2.5 PIV Controller Design with Motor Parameters

The PIV controller is a type of controller used to manage and regulate systems by adjusting the system's behavior based on error signals. The controller combines Proportional (P), Integral (I), and Velocity (V) components to provide effective and adaptive control in systems with dynamic behaviors.

Block reduction of PIV controller:

Closed loop transfer function is obtained as following

$$G(s) = \frac{AKps + AKi}{s^3 + s^2(B+AKv) + s(AKp) + AKi} \quad (2.10)$$

By using the pole placement method ended up with this equation:

$$s^3 + s^2(B + AKv) + s(AKp) + AKi = (s + a)^3 \quad (2.11)$$

by substituting A and B values and comparing the two sides, these values for a , and Kp, Kv, Ki are reached:

$$Ki = \frac{a^3}{A} \quad (2.12)$$

$$Kp = \frac{3a^2}{A} \quad (2.13)$$

$$Kv = \frac{3a - B}{A} \quad (2.14)$$

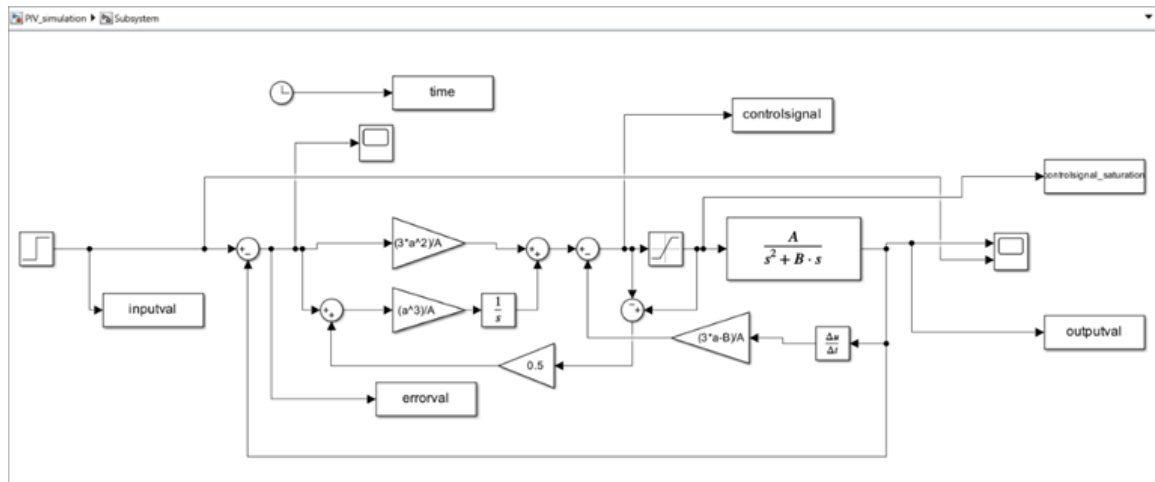


Figure 2.11: Block Diagram of PIV Controller

Wind-up happens when the integral term in a controller continues to accumulate error even though the actuator is already saturated. In the control system here, anti-windup compensation with saturation block is applied to prevent integrator wind-up, which can occur when the controller output exceeds actuator limits.

The difference between the unsaturated and saturated control signals is computed and scaled using a feedback gain. This correction signal is fed back into the integrator path to counteract the excessive integration that would occur during saturation. This method ensures that the integrator does not accumulate error.

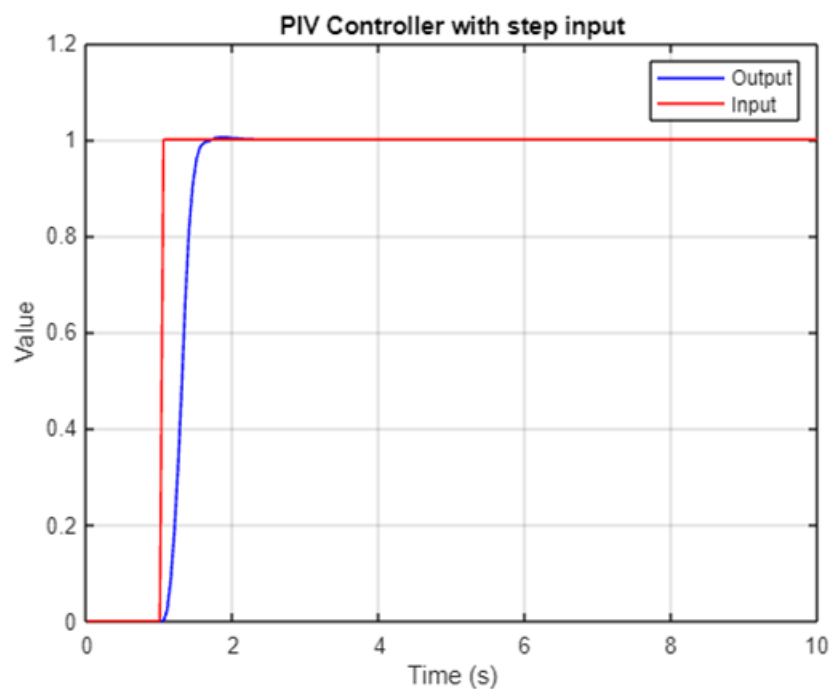


Figure 2.12: Input Comparison with PIV Controller

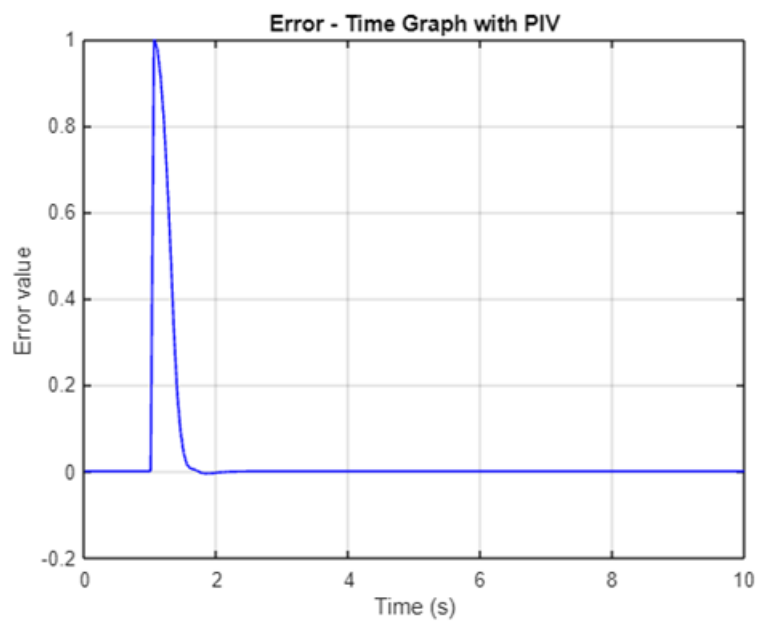


Figure 2.13: Error result from PIV Controller

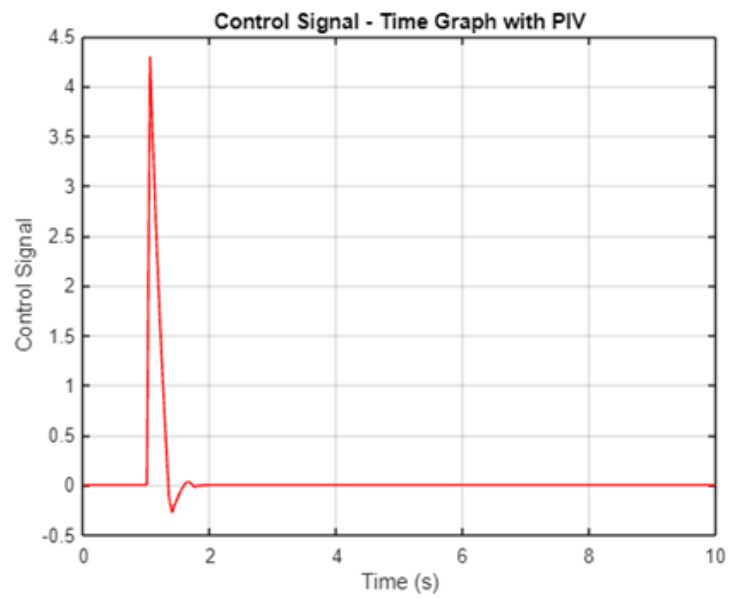


Figure 2.14: Control Signal in PIV controller

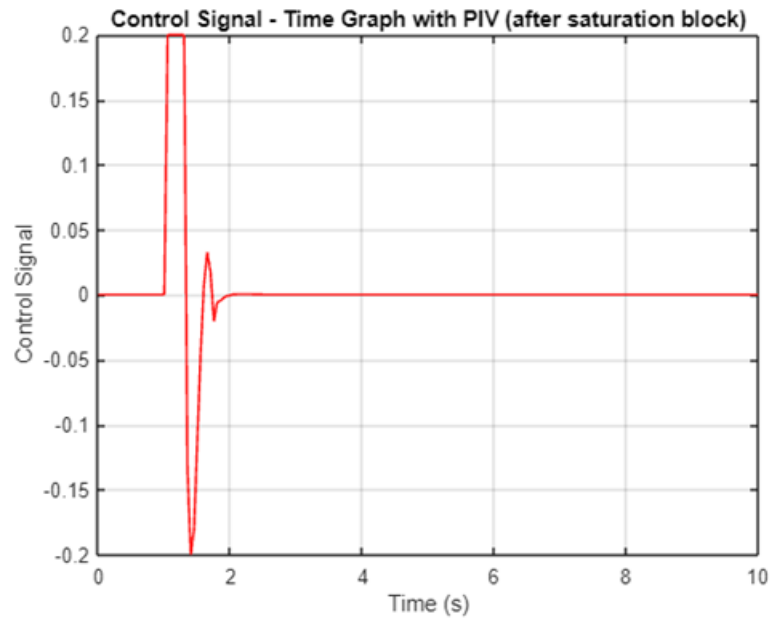


Figure 2.15: Control signal after using saturation block

2.6 Electronic Circuit

Designing the electronic circuit, it is aimed to control the position and speed of two Pololu DC motor with encoder using an Arduino Mega2560 and an L298N dual H-bridge motor driver module.

Table 2.1 Motor Driver and Arduino Pins

L298N Pin	Description	Arduino Pin
IN1	Motor direction control	Pin 5
IN2	Motor direction control	Pin 6
ENA	Motor speed control (PWM)	Pin 9 (PWM)
GND	Ground	GND
VCC	Motor power (12V input)	External 12V power supply
OUT1/OUT2	Motor terminals	Connected to motor

Table 2.2 Encoder Wires and Arduino Pins

Encoder Wire	Function	Arduino Pin
A (Channel A)	Encoder signal	Pin3(Interrupt)
B (Channel B)	Encoder signal	Pin 2
VCC	Encoder power	External 12V
GND	Encoder ground	GND

2.6.1 PCB Design

The electronic control board was designed using KiCad. This PCB is responsible for driving the DC motors and incorporates key components, including a voltage regulator (LM1117), a motor driver (DRV8871DDA), and a microcontroller (STM32G051K6T6). The STM32 microcontroller generates PWM signals, which are used to control the operation of the DC motor.

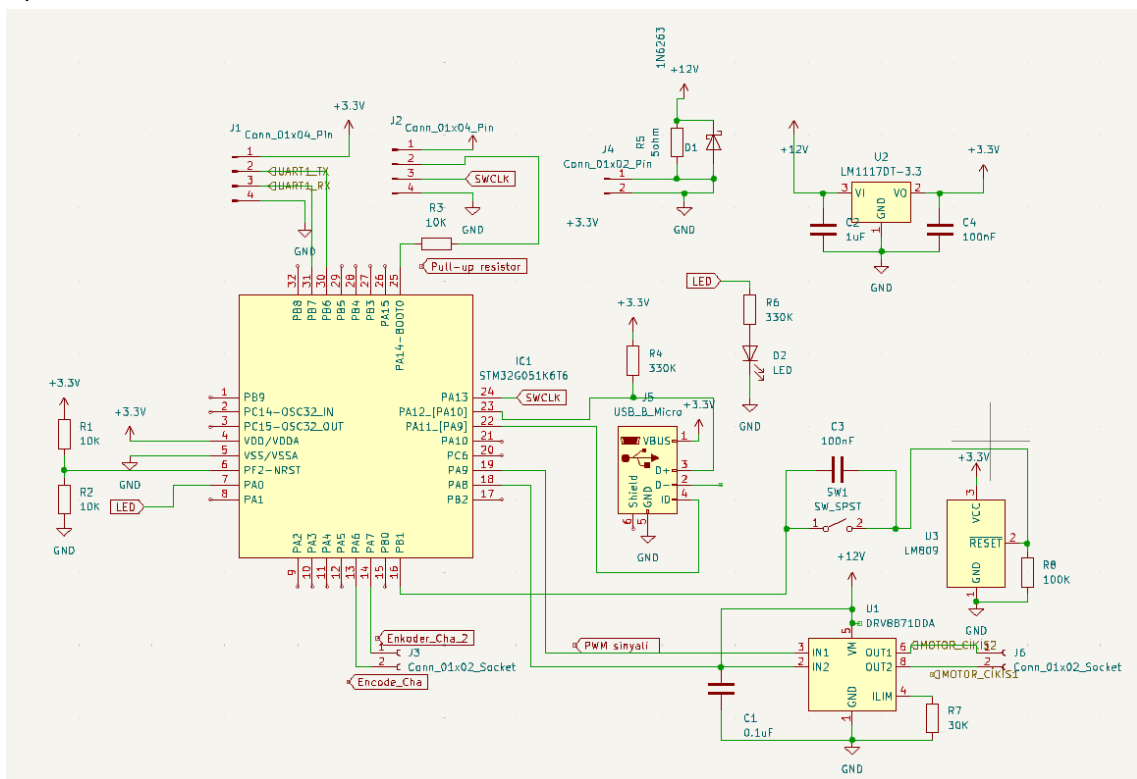


Figure 2.16: Schematic Circuit in KiCad

The DRV8871DDA is an H-bridge motor driver that allows for bidirectional control of DC motors. It uses four N-channel MOSFETs to regulate the current flow through the motor. The motor's speed and direction are controlled via PWM signals applied to the gate terminals of these MOSFETs. A MOSFET becomes conductive when a voltage above a certain threshold is applied between the gate and source terminals. Once turned on, current flows from the drain to the source, enabling motor movement.

The LM1117-3.3V is a linear voltage regulator used to step down the input voltage to a stable 3.3V output, which is necessary for powering low-voltage components such as the STM32 microcontroller



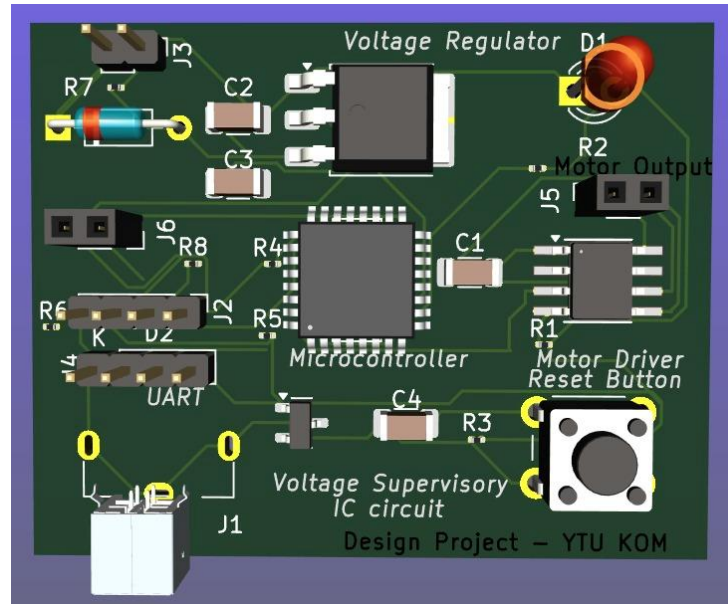


Figure 2.18: PCB Electronic Card Design in KiCAD

2.7 Mechanical Design and Implementation

The mechanical design of the shaking table system was developed by considering dynamic stability, structural rigidity, and ease of manufacturing. The fundamental design comprises a sturdy base, two orthogonal moving stages, and the necessary drive systems to realize X and Y axis motions.

The base frame of the shaking table was constructed using 400 x 400 mm sigma profile aluminum, chosen for its modular design, high strength-to-weight ratio, and ease of assembly. The table consists of two moving stages. Both the middle and upper layers were manufactured using 360 x 360 mm aluminum plates to ensure lightweight construction and minimize the overall inertia of the system.

The actuation system is based on two SCR1610 ball screws, one for each axis, selected for their high efficiency and precise linear motion. The SCR1610 model features a 16 mm diameter and a 10 mm pitch, ensuring high positioning accuracy and smooth operation. The rotational motion from the motors is converted to linear displacement via these ball screws. Considering the pitch of the ball screw, one full revolution of the motor corresponds to 1 cm movement in the horizontal plane. It is sufficient for seismic simulation. The core relationship between motor torque and axial force is given by [14]:

$$F = \frac{2\pi\eta T}{p} \quad (2.15)$$

where η represents the ball screw efficiency (approximately 0.9), T is the torque applied by the motor (0.833 Nm.), and p is the ball screw pitch (10 mm).

$$F = \frac{2\pi \cdot 0.9 \cdot 0.833}{0.01} \approx \frac{4.71}{0.01} = 471N \quad (2.16)$$

This 471 N is the maximum horizontal force the system can apply in the X or Y direction (per axis).

With the Pololu 19:1 Metal Gearmotor 37Dx68L mm 12V motors used in this system, the torque capacity is sufficient to generate the required forces for seismic simulation, while maintaining reliable performance under load.

For the linear guidance system, two 16 mm diameter steel shafts were installed per axis. Each shaft is supported by SCE16UU linear bearings (two per shaft) to minimize friction and ensure smooth motion under dynamic conditions. This configuration provides the necessary support to counteract lateral loads during table acceleration. Eight shaft supports were installed to secure the linear shafts rigidly to the base frame and moving stages.

To mount the ball screws, each axis utilizes one BK12 and one BF12 support, ensuring proper alignment and handling of axial and radial loads. Additionally, an SFU1610 ball screw nut with an appropriate nut housing was used per axis to securely transfer loads between the ball screw and the moving stage. To connect the motor shafts to the ball screws, 6x10 mm couplings were installed, which accommodate minor misalignments and minimize vibrations.

A critical design consideration was the mounting of the motors. For the motor on the bottom, a 3D-printed fixture was designed and attached to the sigma profile base. This custom part, manufactured from high-strength PLA material, hence the motor is securely fixed.

A complete 3D model of the entire system was developed using SolidWorks, providing a comprehensive layout of all components and their spatial relationships. This model was used to verify the mechanical integration of each subsystem, ensuring that the ball screws, linear bearings, and support structures were properly aligned, and that no collisions would occur during full-range motion.

The SolidWorks model and actual hardware of the system are shown in the figures below.

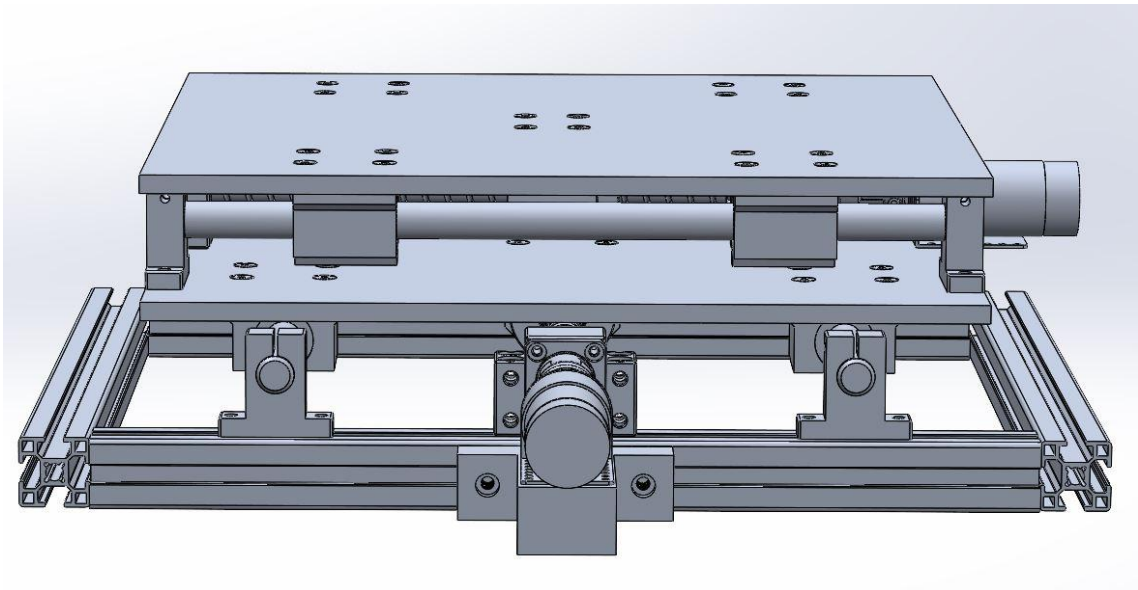


Figure 2.19: SolidWorks Design of the Shaking Table Side View

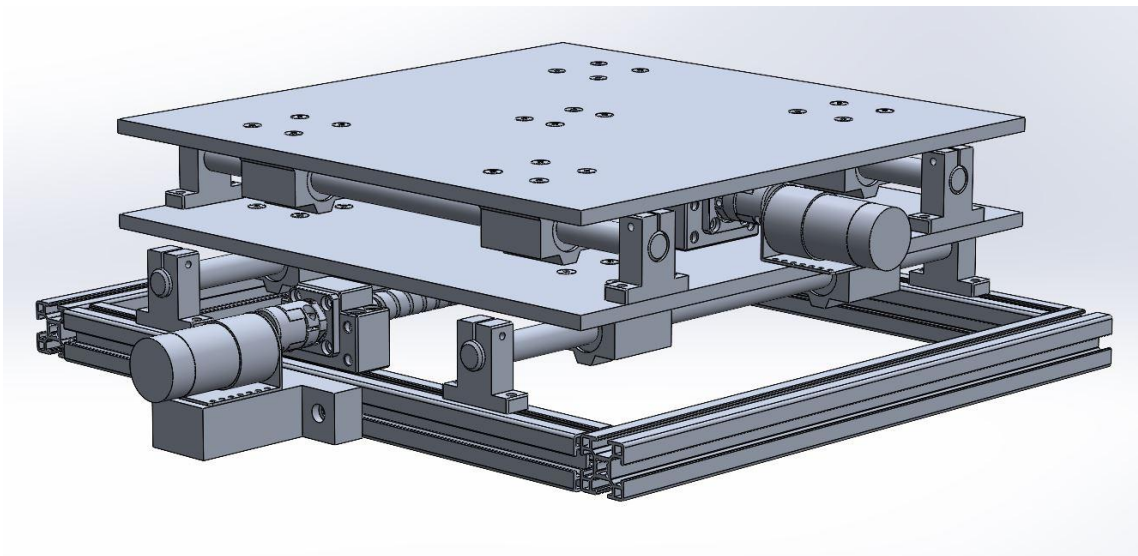


Figure 2.20: SolidWorks Design of the Shaking Table Diagonal View

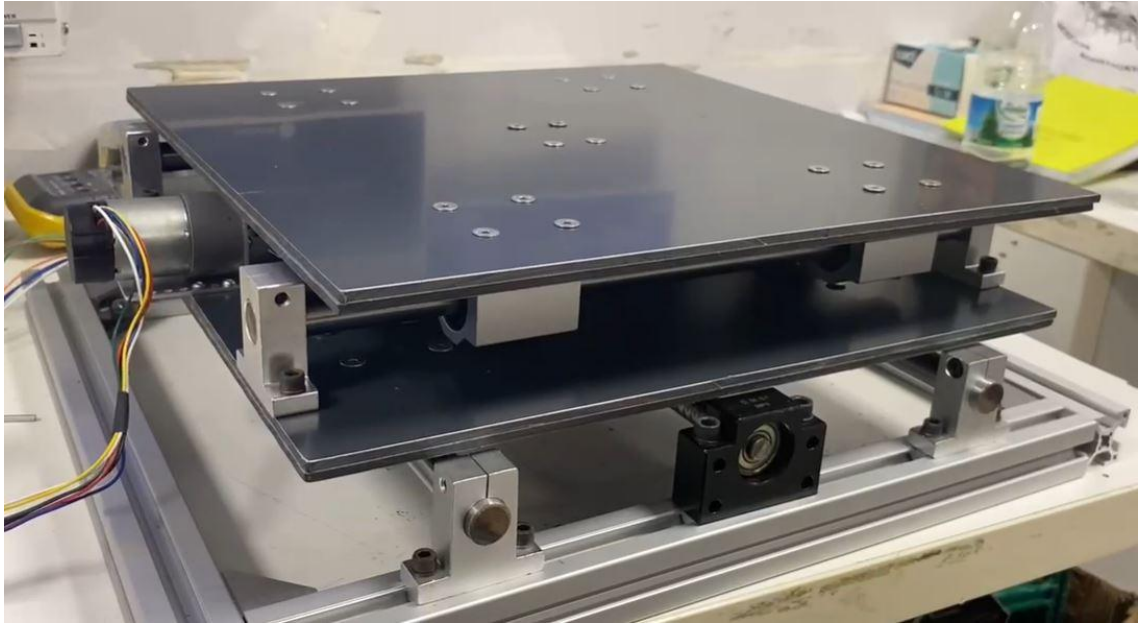


Figure 2.21: The Hardware of the Shaking Table

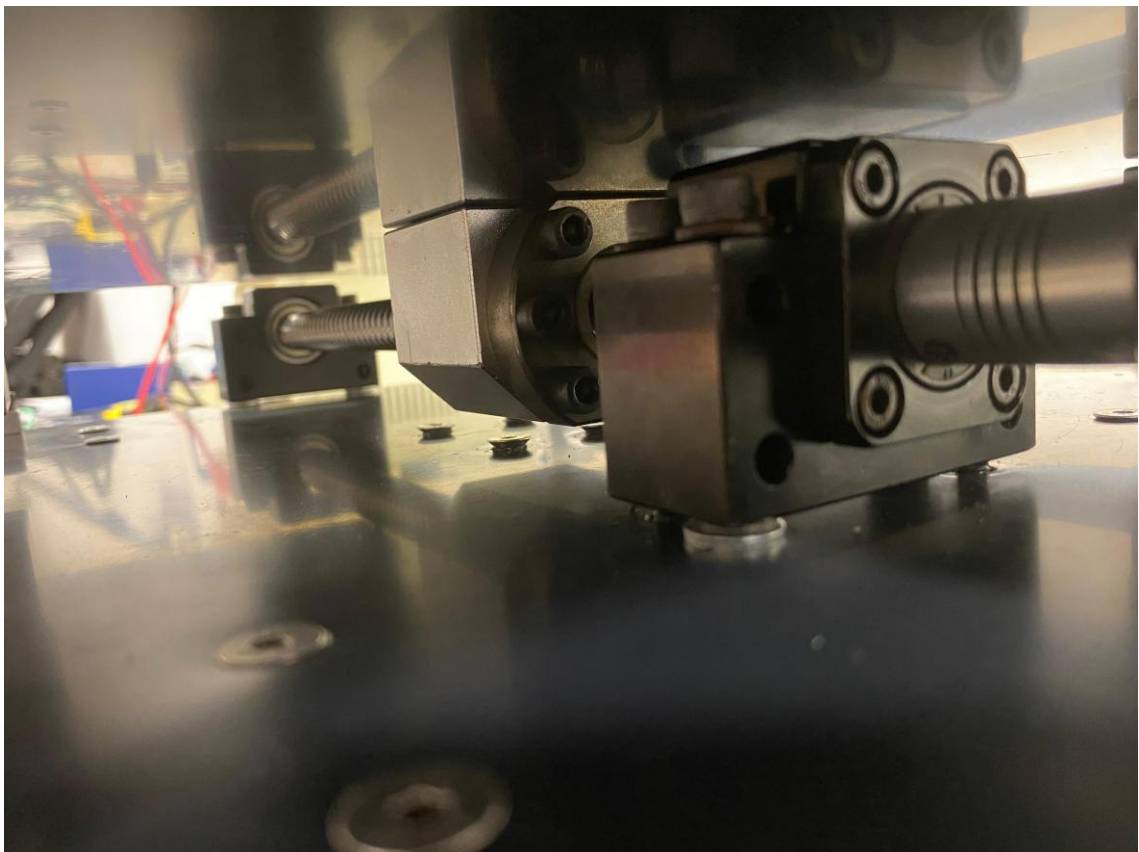


Figure 2.22: The Ball Screw System of Shaking Table

REALISTIC LIMITS, CONDITIONS AND CONSTRAINS TAKEN INTO CONSIDERATION IN THE DESIGN OF THE PROJECT

In the design of the shaking table, several practical limits, conditions, and constraints were carefully considered to ensure functionality, reliability, and safety. These include:

- **Physical Limitations, Constraints, and Boundaries:**

The 2-DOF shaking table is designed considering mechanical constraints such as the maximum payload capacity and structural stiffness to ensure stability during operation. Electrical limitations include the power ratings of the actuators and controllers, which limit the maximum achievable acceleration and velocity. Mechanical factors like friction in joints, actuator saturation, and hysteresis in sensors are taken into account to improve the accuracy and repeatability of the system. The working space and movement trajectories are limited by the table size and actuator ranges, and physical barriers are considered to prevent collisions or over-travel.

- **Engineering Norms and Standards:**

The design complies with relevant engineering standards for laboratory testing equipment and vibration simulators. Safety regulations regarding electrical components and mechanical design have been applied. Additionally, considerations are made for the real-world applicability of the shaking table in earthquake simulation scenarios.

- **Economic aspects:**

The project aims to provide a cost-effective solution for seismic testing by using affordable components while maintaining adequate performance. A detailed bill of materials has been prepared, including costs for actuators, sensors, structural materials, and control electronics, to ensure the project remains within budget constraints. The design could benefit research institutions and universities by providing an accessible testing platform.

- **Environmental Aspects:**

The shaking table is designed to minimize energy consumption during operation.

- **Sustainability:**

The 2 DOF shaking table supports sustainable development by enabling more accurate earthquake simulations, which can help improve building designs and disaster preparedness. This contributes to safer infrastructures and reduces long-term economic and human losses in the community and country.

- **Manufacturability:**

The design emphasizes manufacturability using commonly available parts and straightforward assembly procedures, facilitating easy production and repair in typical laboratory or workshop environments.

- Health aspects:

The design emphasizes manufacturability using commonly available parts and straightforward assembly procedures, facilitating easy production and repair in typical laboratory or workshop environments.

- Security aspects:

Security measures are implemented to ensure safe operation, including limit switches and software constraints to avoid excessive movements that could damage the system or endanger users

- Ethical aspects:

All data and test results were collected following ethical research principles. Proper legal permissions were obtained, and all sources were accurately cited. No data fabrication or falsification occurred during the study.

RESULTS AND DISCUSSIONS

3.1 Simulink and Output Results

The PID and MRAC + PID models are tested in both bottom and upper motors of the system. Simulink interfaces; output, error, and control signal results are added below.

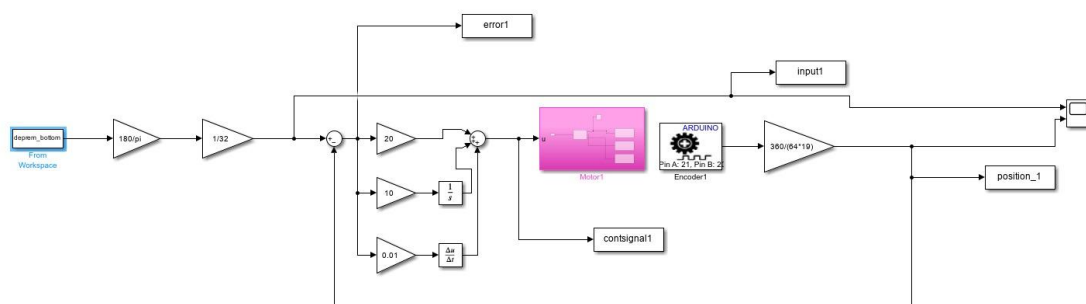


Figure 3.1: PID Bottom Motor Simulink

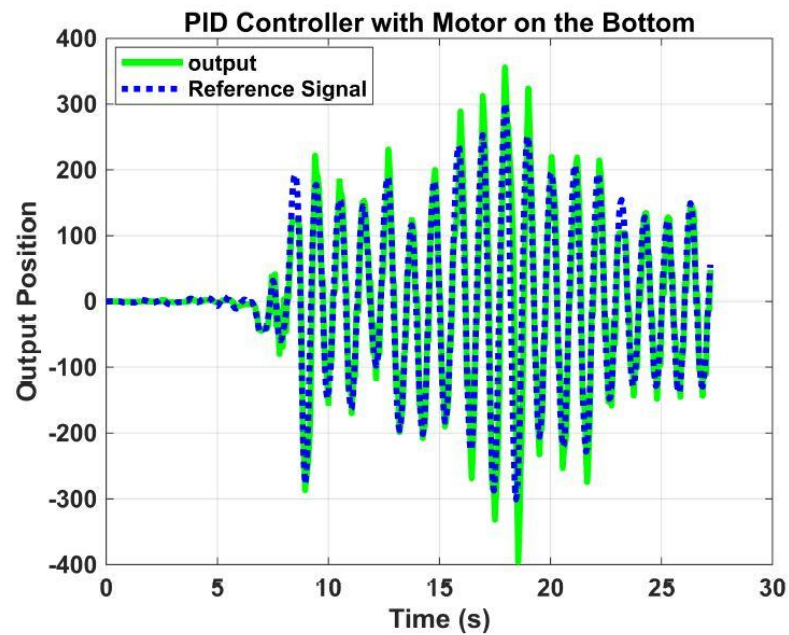


Figure 3.2: PID Bottom Motor Output

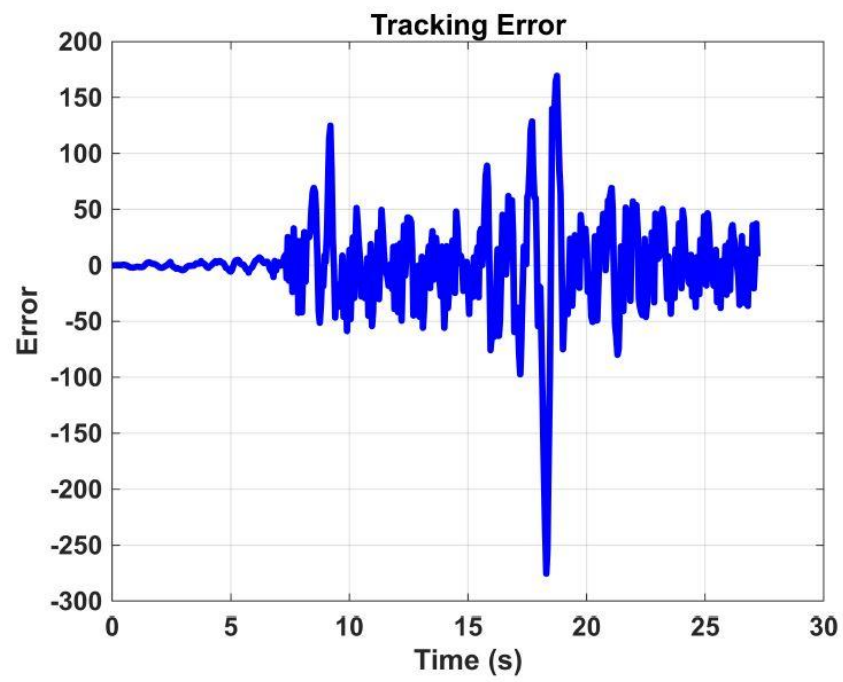


Figure 3.3: PID Bottom Motor Error

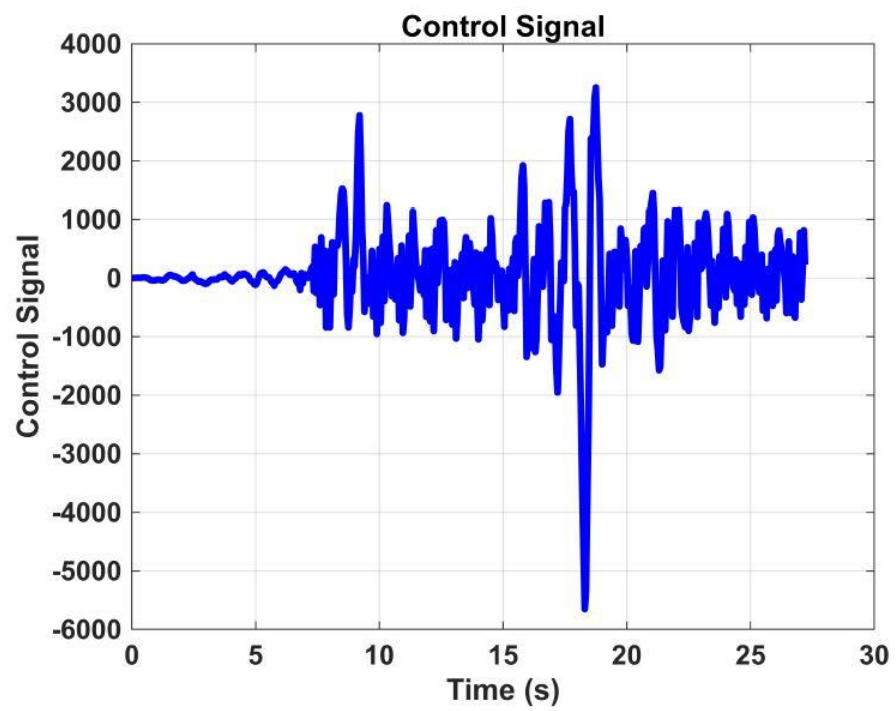


Figure 3.4: PID Bottom Motor Control Signal

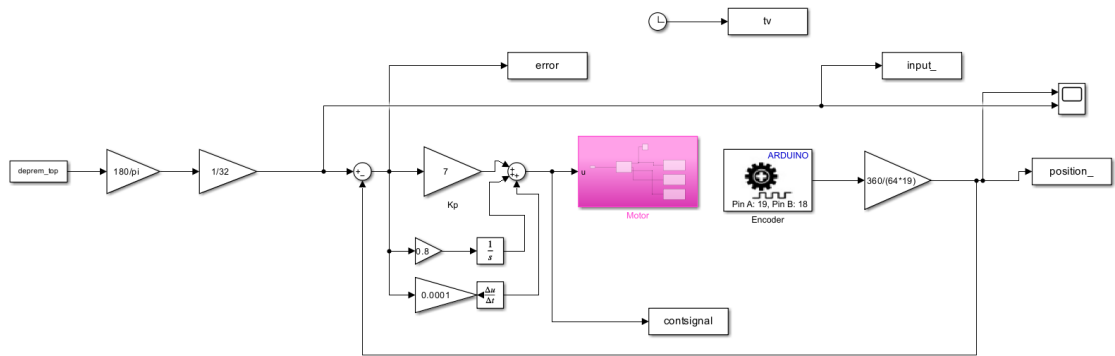


Figure 3.5: PID Top Motor Simulink

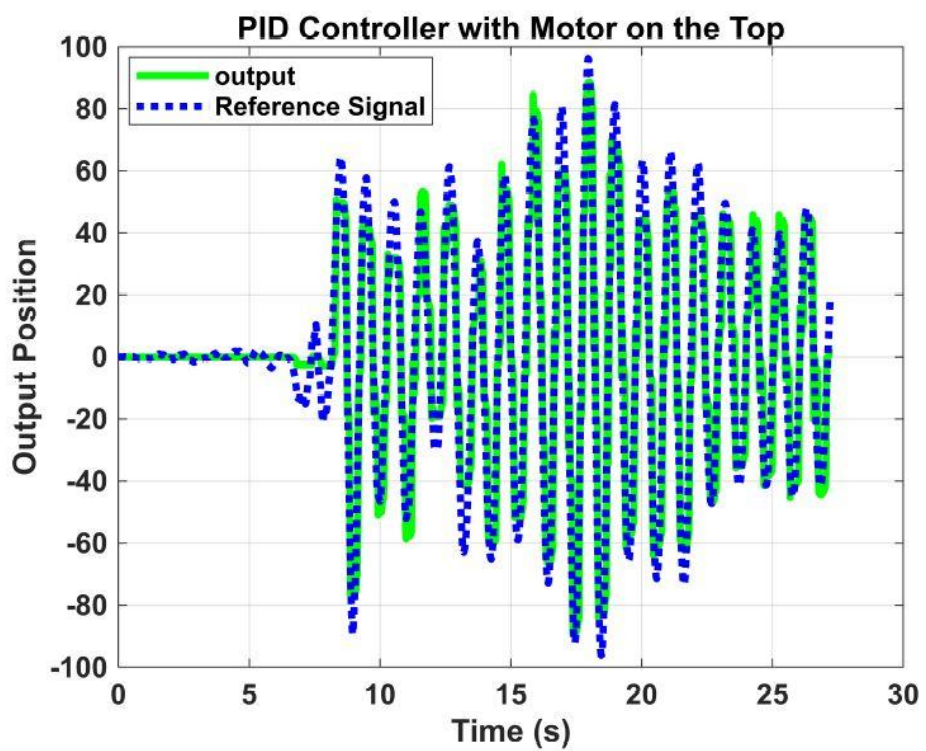


Figure 3.6: PID Top Motor Output

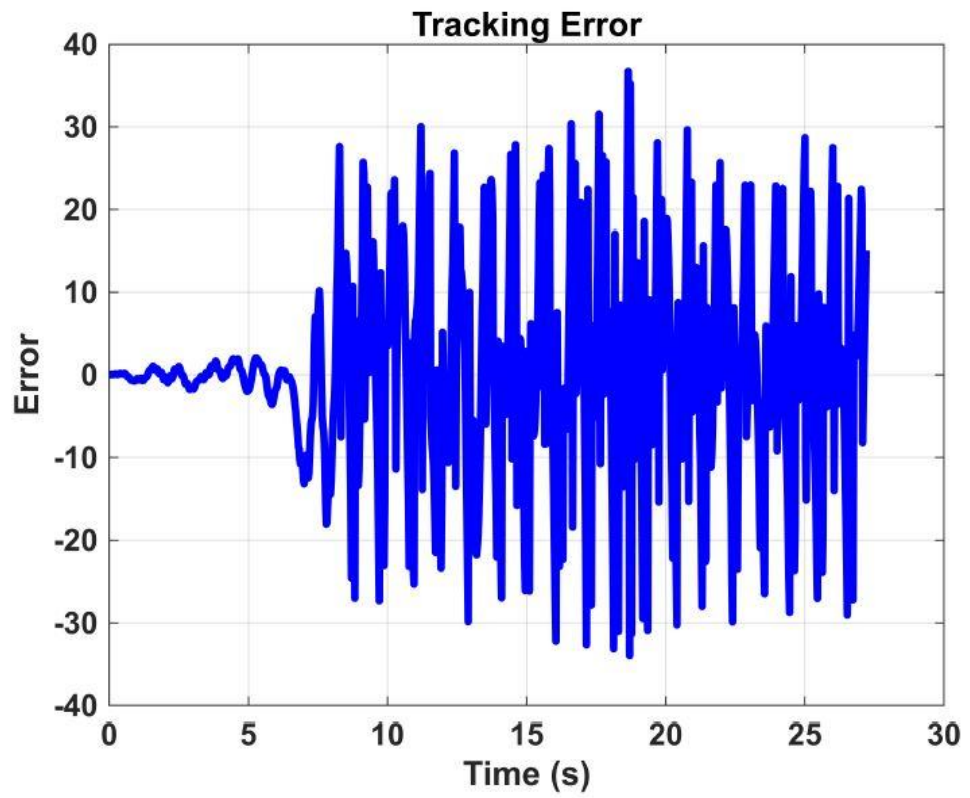


Figure 3.7: PID Top Motor Error

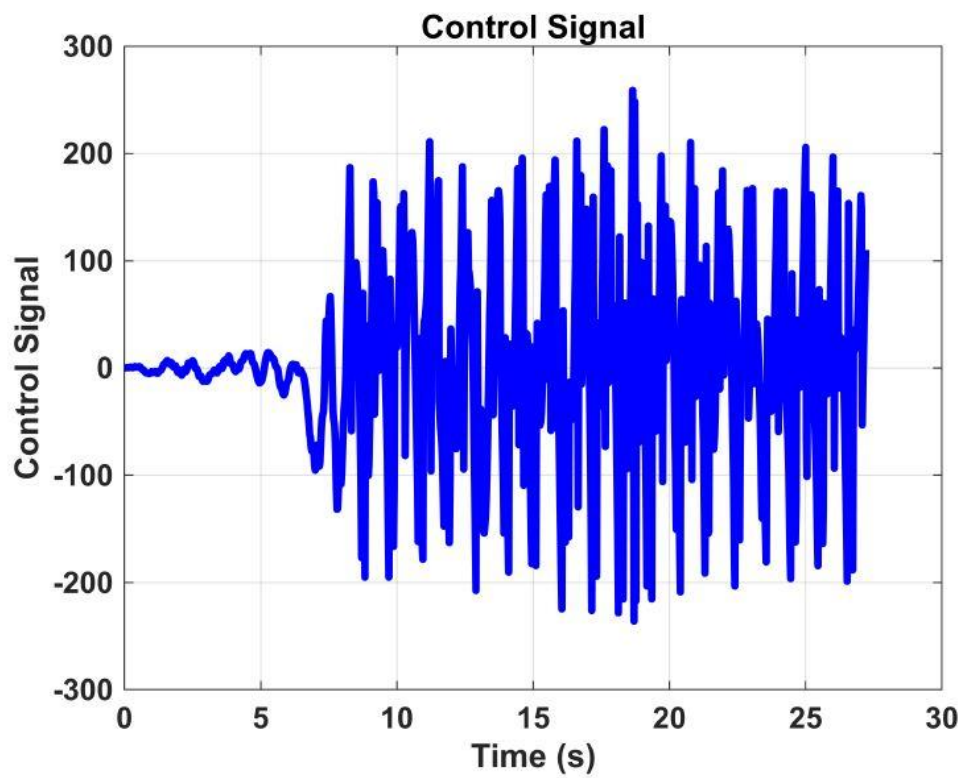


Figure 3.8: PID Top Motor Control Signal

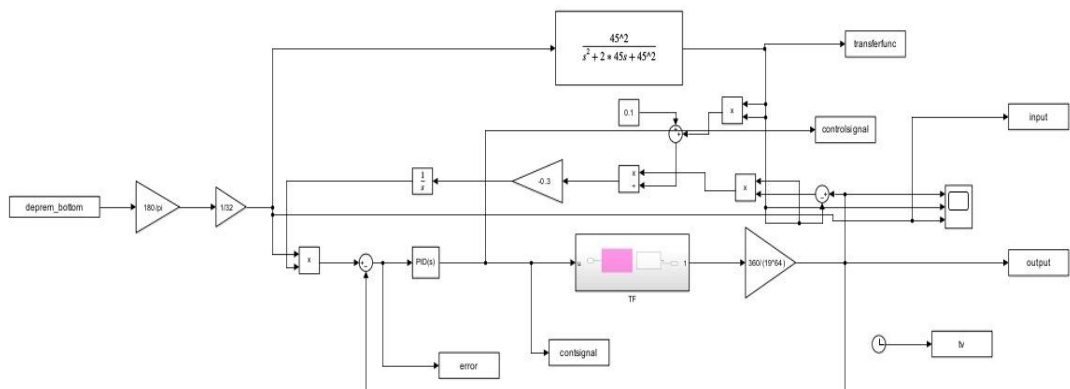


Figure 3.9: MRAC + PID Bottom Motor Simulink

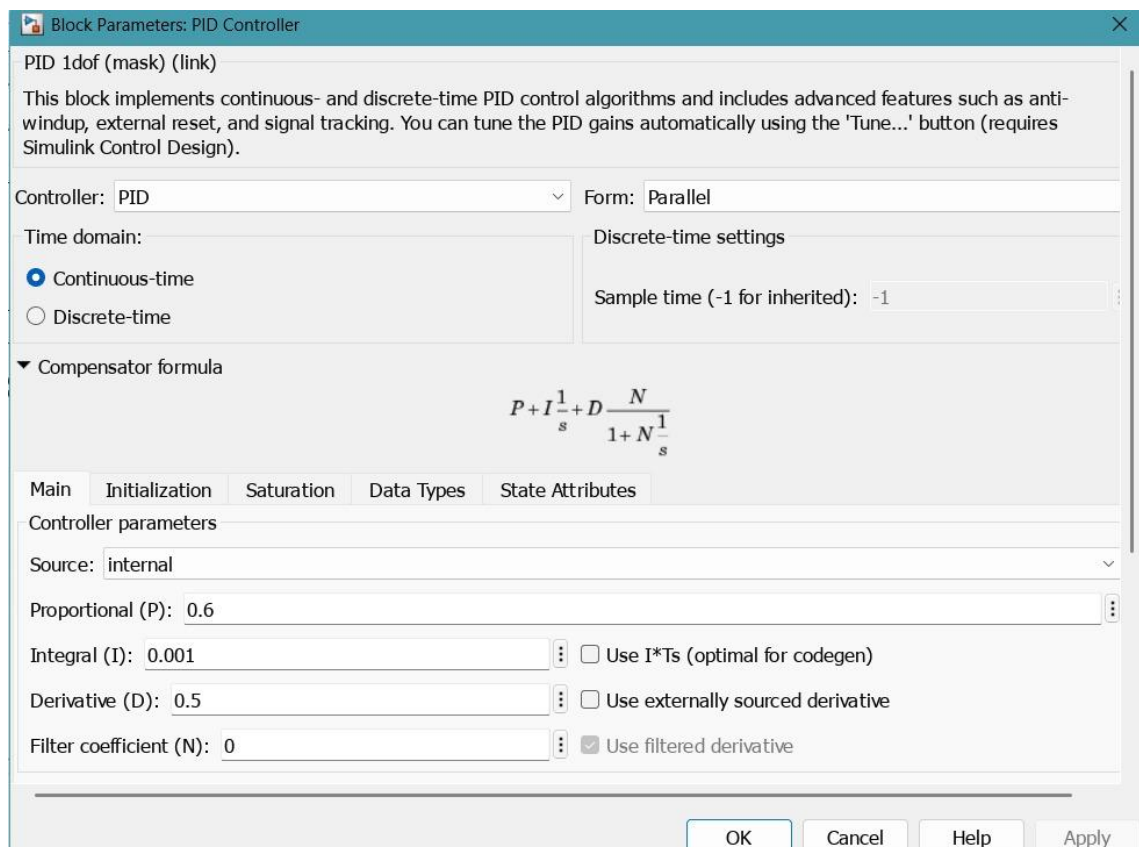


Figure 3.10: MRAC + PID Bottom Motor PID Gains

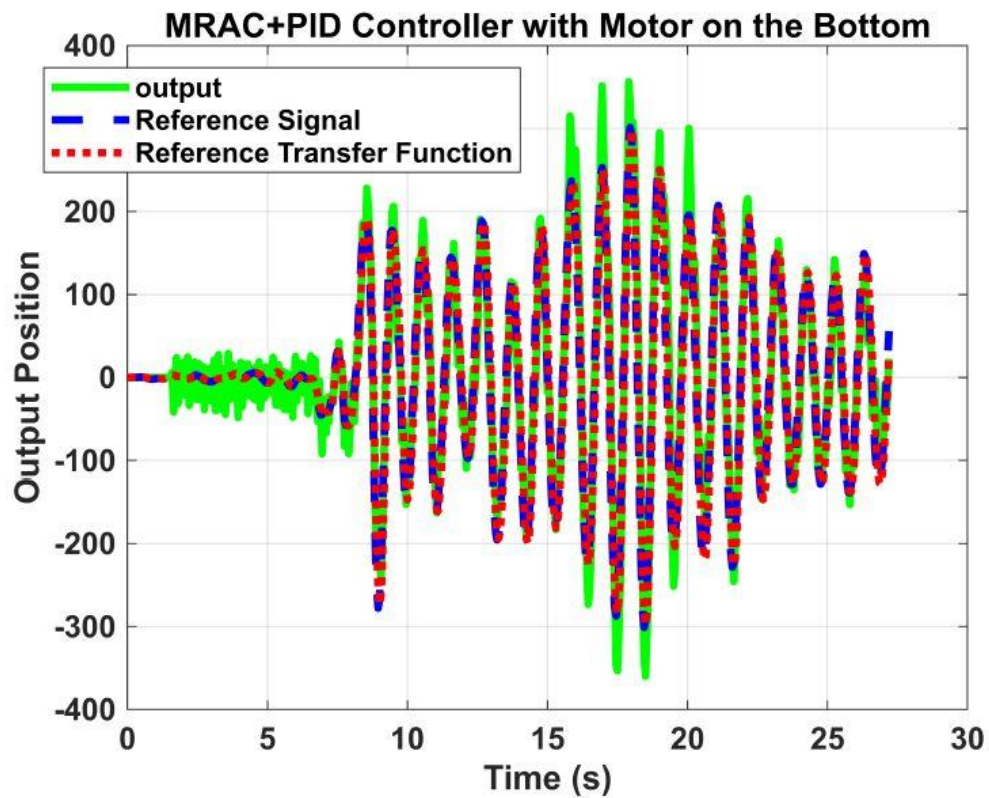


Figure 3.11: MRAC + PID Bottom Motor Output

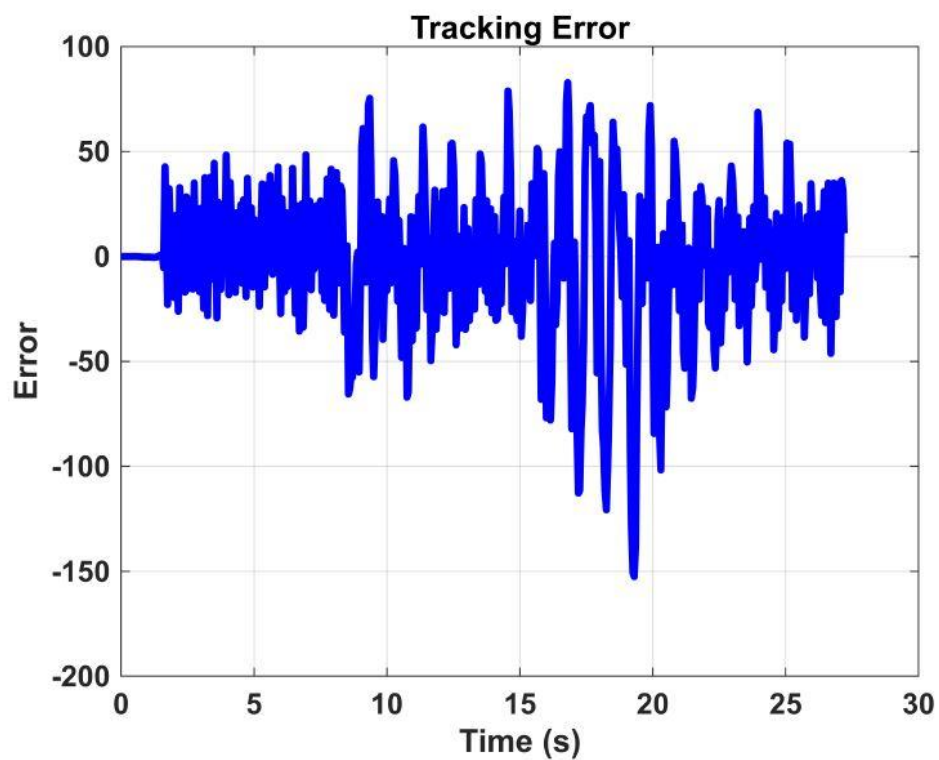


Figure 3.12: MRAC + PID Bottom Motor Error

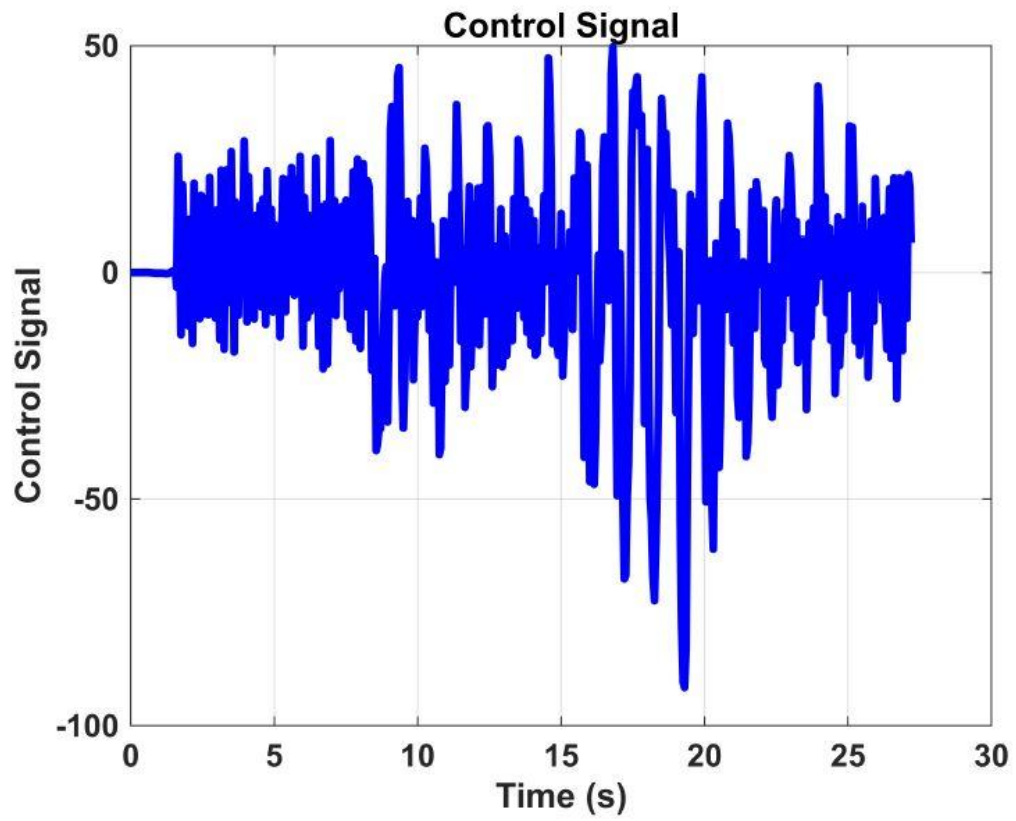


Figure 3.13: MRAC + PID Bottom Motor Control Signal

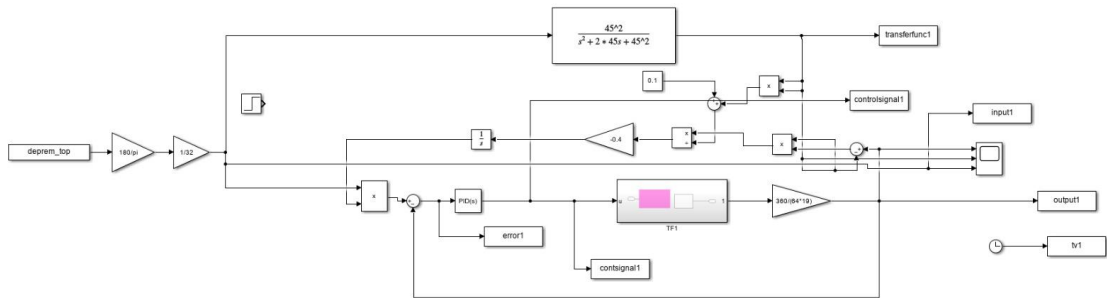


Figure 3.14: MRAC + PID Top Motor Simulink

Block Parameters: PID Controller1

PID 1dof (mask) (link)

This block implements continuous- and discrete-time PID control algorithms and includes advanced features such as anti-windup, external reset, and signal tracking. You can tune the PID gains automatically using the 'Tune...' button (requires Simulink Control Design).

Controller: PID Form: Parallel

Time domain:

☒ Continuous-time

☐ Discrete-time

Discrete-time settings

Sample time (-1 for inherited): -1

Compensator formula

$$P + I \frac{1}{s} + D \frac{N}{1 + N \frac{1}{s}}$$

Main Initialization Saturation Data Types State Attributes

Controller parameters

Source: internal

Proportional (P): 6

Integral (I): 0.01 ☐ Use I*Ts (optimal for codegen)

Derivative (D): 0.3 ☐ Use externally sourced derivative

Filter coefficient (N): 0 ☒ Use filtered derivative

Automated tuning

OK Cancel Help Apply

Figure 3.15: MRAC + PID Top Motor PID gains

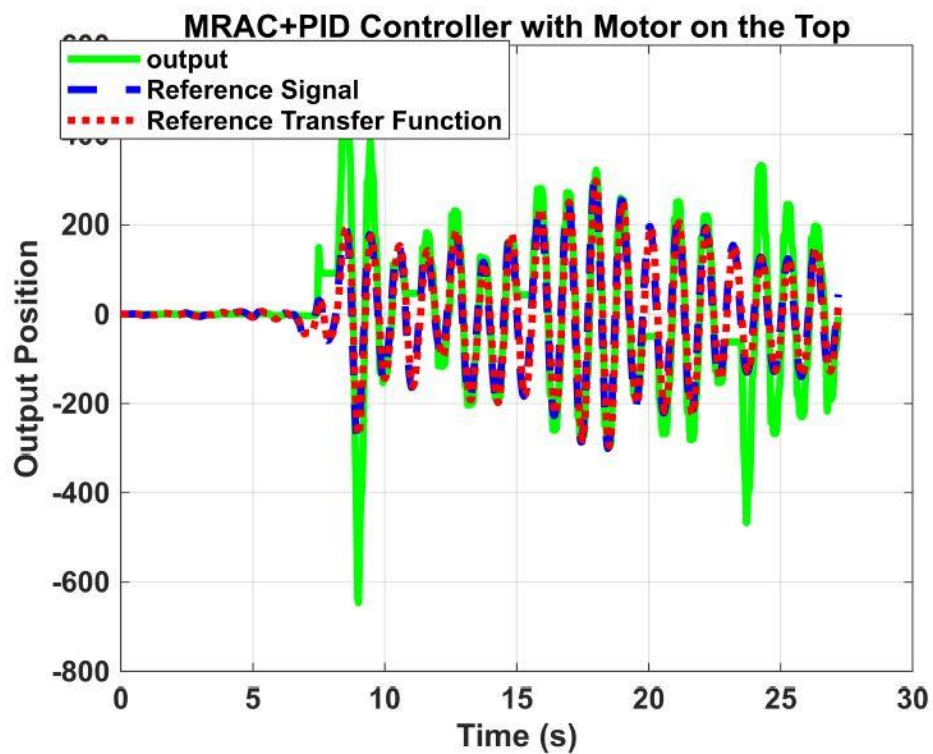


Figure 3.16: MRAC + PID Top Motor Output

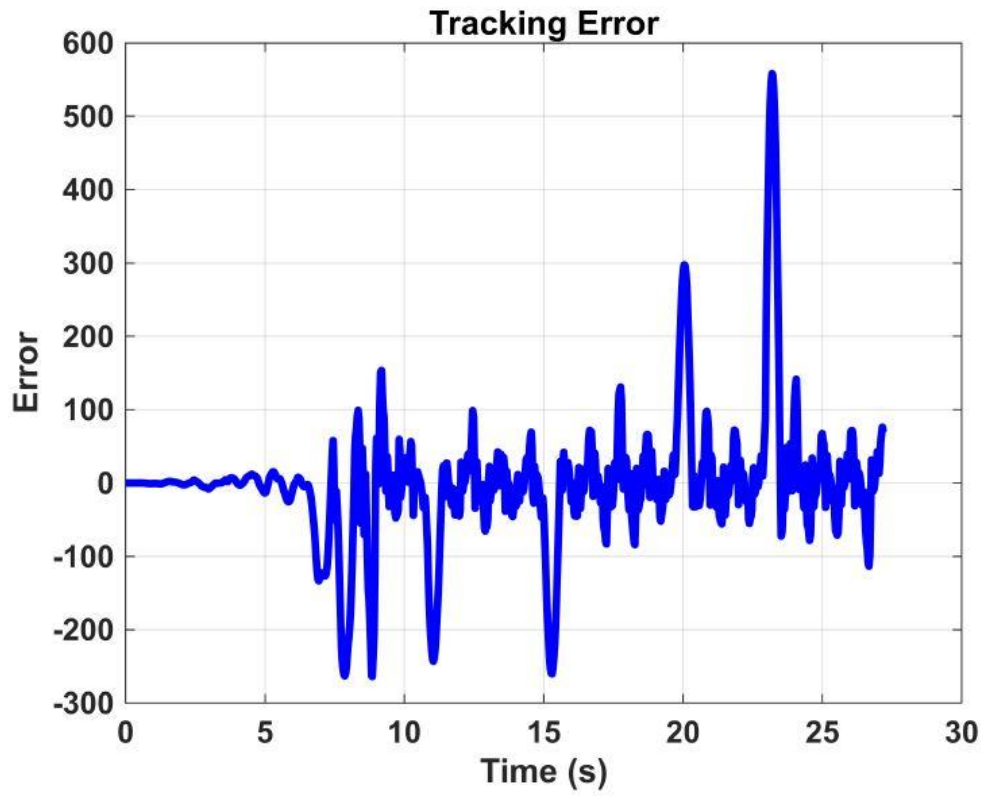


Figure 3.17: MRAC + PID Top Motor Error

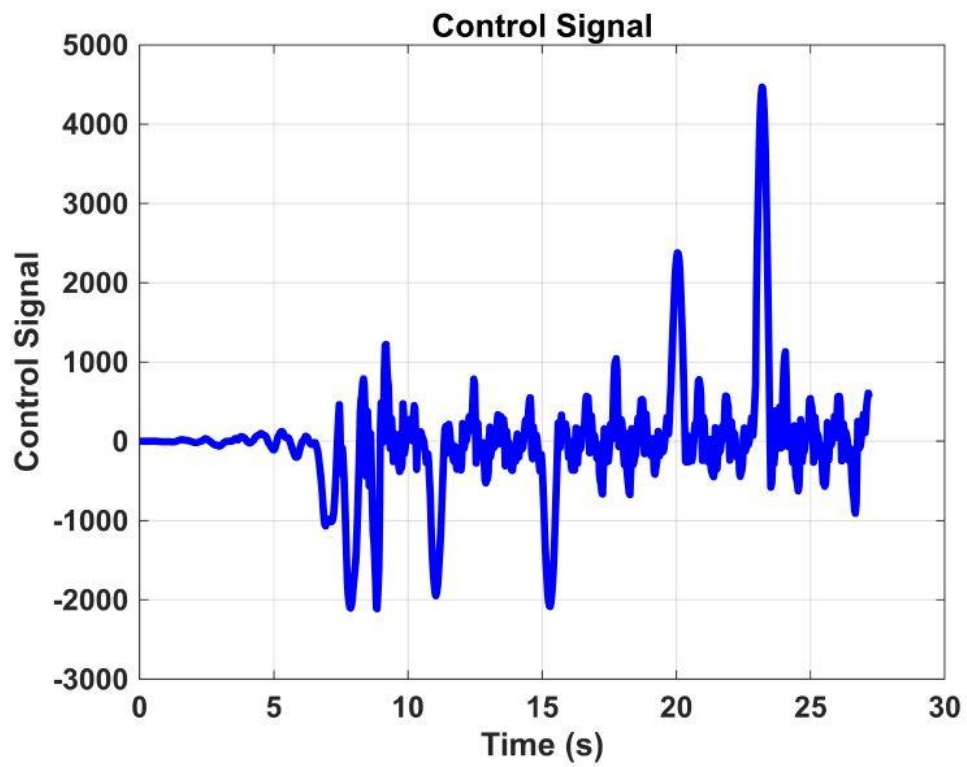


Figure 3.18: MRAC + PID Top Motor Control Signal

3.2 Comparison of the Bottom Motor's PID and MRAC + PID Results

The performance of MRAC+PID and classical PID controllers are evaluated for the configuration where the motor is mounted on the bottom platform. Both controllers were tested under the same reference signal, and their tracking accuracy and error magnitudes were compared.

Figure 3.11 and Figure 3.12 present the results of the MRAC+PID control structure. Thanks to its adaptive nature, the MRAC controller successfully tracked the reference transfer function with high accuracy. Even during changes in system dynamics, a strong correlation was maintained between the system output and the reference model. The tracking error plot (Figure 3.12) shows that although the error increases during certain intervals, the system remains stable and continues to follow the desired trajectory.

However, Figure 3.2 and Figure 3.3 show the performance of the classical PID controller. While the PID controller successfully tracks low-frequency, small-amplitude reference inputs, it starts to deviate from the desired signal as the reference becomes more dynamic. The tracking error plot (Figure 3.3) reveals sudden deviations and sharp spikes in error, indicating that the PID controller struggles to adapt to rapid changes in system behavior.

Overall, the MRAC+PID control strategy provides better tracking accuracy than the classical PID controller. Its adaptive structure allows it to better handle nonlinear effects and dynamic variations, resulting in superior performance in reference model tracking.

3.3 Comparison of the Top Motor's PID and MRAC + PID Results

The control performance is evaluated for the configuration in which the motor is placed on the top platform of the 2-DOF shaking table. Both classical PID and MRAC+PID controllers were tested under the same input conditions, and their tracking capabilities were compared.

Figure 3.16 and Figure 3.17 illustrate the performance of the MRAC+PID controller. Although the system experiences a brief instability during the initial excitation period (around 8–10 seconds), the MRAC+PID controller effectively stabilizes the system and achieves a close match with both the reference signal and the reference model. The reference tracking error, shown in Figure 3.17, demonstrates large initial deviations but maintains bounded error over time despite nonlinear dynamics and disturbances. This result emphasizes the adaptive controller's ability to adjust to changes in system parameters.

On the other hand, Figure 3.6 and 3.7 present the results obtained with the classical PID controller. The system successfully tracks the reference signal during lower amplitude phases; however, it begins to deviate significantly during high-frequency and high-amplitude input sections. The tracking error remains bounded (within ± 30)

but exhibits persistent oscillations and a lack of adaptation, especially under varying load dynamics.

Compared to the PID controller, the MRAC+PID structure offers superior flexibility in adapting to the nonlinearities and mechanical complexities introduced by the top-mounted motor configuration. Although the adaptive controller experienced large momentary tracking errors, it was more successful overall in mimicking the desired reference model behavior throughout the test duration.

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MATLAB CODES

To characterize the motor's steady-state behavior, PWM inputs were applied from -1 to 1 in steps of 0.05 using Arduino Mega2560. For each PWM command, the motor speed was measured using a rotary encoder. The raw data was then filtered to remove non-monotonic points, allowing the creation of a clean 1D lookup table.

```
clear all
clc
close all

%% 1. Create test data
maxPWM = 1.00;
incrPWM = 0.05;
PWMcmdRaw = (-maxPWM:incrPWM:maxPWM)';

%% 2. Create and initialize device objects
clear a encoderObj
a = arduino('COM7', 'Mega2560', 'Libraries', 'RotaryEncoder'); % COM portunu
gerektiği gibi değiştir

% Motor pins
in1 = 'D7'; % Motor direction pin1
in2 = 'D8'; % Motor direction pin2
ena = 'D6'; % PWM pin

configurePin(a, in1, 'DigitalOutput');
configurePin(a, in2, 'DigitalOutput');
configurePin(a, ena, 'PWM');

% Encoder pins
encoderObj = rotaryEncoder(a, 'D3', 'D2');
resetCount(encoderObj);

%% 3. Measure raw motor speed for each PWM command
speedRaw = zeros(size(PWMcmdRaw)); % Pre-allocate vector

for ii = 1:length(PWMcmdRaw)
    resetCount(encoderObj);
    pwmVal = abs(PWMcmdRaw(ii));

    % checking the direction of the motor
    if PWMcmdRaw(ii) >= 0
        writeDigitalPin(a, in1, 1);
        writeDigitalPin(a, in2, 0);
    else
        writeDigitalPin(a, in1, 0);
        writeDigitalPin(a, in2, 1);
    end

    tic;
    writePWMPWM(a, ena, pwmVal * 5); %PWM voltage between 0-5V
    pause(1); % wait for steady state
    dt = toc;

    count = readCount(encoderObj);
    speedRaw(ii) = count / dt;
end
```

```

        writePWMMVoltage(a, ena, 0); % Stopping the motor
        pause(1);
    end

    %% 4. Graph raw data
    figure;
    plot(PWMCmdRaw, speedRaw, 'b.-');
    title('100:1 Gearbox Motor Steady State Response')
    xlabel('PWM Command')
    ylabel('Measured Speed (counts/sec)')

    %% 5. Post-process and save data
    idx = (diff(speedRaw) > 0);
    speedMono = speedRaw(idx);
    PWMCmdMono = PWMCmdRaw(idx);
    PWMCmdMono(speedMono == 0) = 0;
    save motorResponse PWMCmdMono speedMono

    %% 6. Graph raw and post-processed data
    figure;
    plot(PWMCmdRaw, speedRaw, 'b.-');
    hold on;
    plot(PWMCmdMono, speedMono, 'ro-');
    title('100:1 Gearbox Motor Steady State Response')
    xlabel('PWM Command')
    ylabel('Measured Speed (counts/sec)')
    legend('Raw Data', 'Monotonic Data')

    %% 7. Save file
    save(mfilename, 'PWMCmdMono', 'speedMono')

    %% 8. Delete device objects
    clear a encoderObj

```

```

function [IN_PWM, INA1, INA2, PWM_EN] = fcn(u)

PWM_EN = u;
IN_PWM=abs(u);
if u<0
    INA1=0;
    INA2=1;
elseif u>0
    INA1=1;
    INA2=0;
else
    INA1=0;
    INA2=0;
end

```

RESUME

1. STUDENT PERSONAL INFORMATION

Name SURNAME : Aylin ÖZTÜRK
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EDUCATION STATUS

Degree	Field	School/University	Graduation Year
High School	Science and Math	Toki Cumhuriyet Anatolian	2020
Bachelor of Science	Control and Automation Engineering	High School Yildiz Technical University	2025

JOB EXPERIENCES

Year	Firm	Duty
2024	Airties Wireless Communication	Test Automation Engineering Intern
2024	Metro Istanbul Rail Systems Management R&D (Electrical Systems)	Engineering Intern
2024	Abdi Ibrahim Pharmaceuticals	Long Term Maintenance & Repair Intern

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EDUCATION STATUS

Degree	Field	School/University	Graduation Year
High School	Science and Math	Özel Balıkesir Uğur Anatolian High School	2020
Associate Degree	Computer Programming	Istanbul University, Faculty of Open and Distance Education	2024
Bachelor of Science	Control and Automation Engineering	Yildiz Technical University	2025

JOB EXPERIENCES

Year	Firm	Duty
2024	BEKO	Automation-Maintenance Intern
2024	OYAK Renault	Vehicle Electronics Process Intern
2025	Deftport	Process Automation Intern

3. STUDENT PERSONAL INFORMATION

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EDUCATION STATUS

Degree	Field	School/University	Graduation Year
High School	Science and Math	Özel Mod Eryaman Anatolian High School	2020